

$$\textcircled{1} \quad A \vdash (A \rightarrow B) \vee (B \rightarrow A)$$

$$\frac{\frac{A}{B \rightarrow A}}{(A \rightarrow B) \vee (B \rightarrow A)}$$

$$\textcircled{2} \quad A \vee B \vdash (A \rightarrow B) \vee (B \rightarrow A)$$

$$\begin{array}{ccc} & \frac{[A]}{B \rightarrow A} & \frac{[B]}{A \rightarrow B} \\ A \vee B & \frac{}{(A \rightarrow B) \vee (B \rightarrow A)} & \frac{}{(A \rightarrow B) \vee (B \rightarrow A)} \\ \hline & (A \rightarrow B) \vee (B \rightarrow A) & \end{array}$$

③ $(A \rightarrow B) \rightarrow B, \neg A \vdash B$

$[A] \neg A$

$\perp$

B

$A \rightarrow B$

$(A \rightarrow B) \rightarrow B$

B